

Georgia Institute of Technology

CS 4365/6365: Introduction to Enterprise Computing

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Project Checkpoint 1 Report

Group: 16

Name(s): Neel Bhattachayya

Project Name: ContractLens: A Privacy-Preserving Contract Intelligence Platform

Context and Related Work:

Contract review is a critical part of the business process used in vendor agreements, software licensing agreements, NDAs, employment contracts, and other commercial transactions. Legal teams spend an average of 3.2 hours manually reviewing every single contract, with complex agreements requiring 8-12 hours across multiple different reviewers [1]. This work is essential with research from world commerce and contracting indicating that companies lose roughly 9.2% of annual revenue due to poor contract management practices [2]. Before agreements are executed, legal and business teams must identify important clauses and assess risks which can be a lengthy process.

Commercial platforms such as Ironclad and other Contract Lifecycle Management (CLM) systems demonstrate the demand for AI contractual review to speed up the process. The Contract Understanding Atticus Dataset (CUAD) provides over 13,000 expert annotations across 510 commercial contracts and 41 clause categories which can enable the development and evaluation of contract-analysis systems. Recently the ACORD benchmark has highlighted the importance of contract clause retrieval and search of past cases in legal review.

While recent LLM tools can summarize documents and answer questions, much of the underlying contract-review workflow could benefit from more specialized document understanding capabilities such as validated risk scoring, clause classification, contract comparison, and semantic retrieval of relevant cases. ContractLens aims to address this by building a contract intelligence platform that transforms unstructured contracts into structured contract representations that support clause classification, risk analysis, contract comparison, and question answering. The system will make full use of the legal datasets available while supporting privacy preserving local deployment for companies that cannot share sensitive contracts with external AI services.

1. <https://www.cyberbase.ai/blog/contract-review-roi>
2. <https://stealthagents.com/research/ai-contract-review-automation-statistics-2026>

Project Plan (Plan)

Provide clear understanding of the current state of the respective idea/related work to provide context and the starting point.

The goal of ContractLens is to develop a privacy-preserving contract intelligence platform that assists with first-pass contract review. Rather than functioning just as a general purpose chatbot, the system will transform the unstructured contracts into structured representations that support clause classification, semantic retrieval, contract comparison, risk analysis, and question answering.

The system will allow its users to upload PDFs, DOCX, or text format which will be first parsed and segmented into individual clauses and sections. A clause classification component will be used to identify important contract categories such as confidentiality, termination, liability, and intellectual property. These structures representations will then be used downstream for retrieval, comparison, and risk analysis.

ContractLens will support natural-language interaction through an LLM interface, but the primary functionality will be implemented in specialized backend services mentioned earlier. MCP-style tool integrations will be used to expose these services to the LLM interface.

The project will leverage publicly available legal datasets including the Contract Understanding Atticus Dataset (CUAD) for clause classification and evaluation. Performance will be evaluated using standard metrics such as precision, recall, and F1 score for clause classification, as well as retrieval metrics such as Recall@K and Mean Reciprocal Rank (MRR). More evaluation will be done to focus on the quality of contract comparison results, risk identification, and overall contract review reports.

The final deliverable will be a fully functional contract intelligence platform capable of extracting contract structure, identifying key clauses, comparing contractual language, highlighting potential risks, and generating review reports while supporting local deployment for privacy sensitive environments.

Project Deliverables

Deliverable	Description	Technical Stack
Contract Upload and Structured Review Workspace	Users will upload a contract and they can see a breakdown of the document which includes contract type, key sections, identified clauses, parties, and important metadata. This will	FastAPI, PDF Processing Libraries, Document Parsing Tools

	serve as the foundation to feed into downstream tasks.	
Contract Clause Exploration	Users can browse automatically identified clauses such as confidentiality, liability, termination, indemnification, and intellectual property. The system will highlight relevant section and link them back to the original contract.	LegalBERT, HuggingFace Transformers, CUAD Dataset
Contract comparison assistant	Users can compare contracts with a company template or an older version. The system will identify changes in obligations and terms and present them to the user.	Embedding Models, FAISS/ChromaDB, Retrieval Pipeline
Chatbot interface	Users will be able to discuss with a local LLM using natural language about their contract, receive cited answers, retrieve similar contractual language from the model, and generate executive review reports summarizing key findings and risks.	Local LLM, MCP Tools, RAG

Project Milestones

Show the project plan and the planned work for each teammate and remove this line.

Check point	Milestone	Technical Scope & Deliverables	Work Splitup	Status
Check point 1	Project planning and system design	Finalize the project scope, key deliverables, project plan, and project moat. Create github repository	1 person team	Done

Check point 2	Contract ingestion and clause intelligence	Implement the contract upload and parsing pipeline, structured extraction, and clause visualization. Users should be able to upload and view identified categories.	1 person team	pending
Check point 3	Retrieval, Comparison, and risk analysis	Implement semantic retrieval for comparison, implement pipeline for contract comparison, and clearly design and implement risk reporting backed by supporting evidence.	1 person team	pending
Check point 4	Platform and evaluation	Design and build ContractLens UI. Integrate chatbot for intelligent querying. Evaluate classification and retrieval performance with benchmark datasets.	1 person team	pending

Current Progress Report (Match)

- Provide a report on the work done during the last 2 weeks.
- Compare the work done with the initially planned milestone chart.
- Mention what your planned work is for the next 2 weeks.
- Mention any changes to the original plan due to the findings over the past 2 weeks.

This submission represents the initial planning and proposal checkpoint. During the past two weeks I focused on brainstorming multiple ideas and running them by TAs and the professor at office hours. Potential domains were researched and one was found where the tool has a strong wedge and can provide something unique. The project scope was defined and strong datasets for validation and tuning were researched.

Planned work for the next two weeks includes, establishing the development environment and scaffolding for the project in the GitHub repo, acquiring and preprocessing datasets, and implementing an initial contract ingestion and document parsing pipeline.

Since this is the first checkpoint and project planning stage, no significant changes have been made to the original project plan.

Supporting Evidence (Factual)

- Add the URL to your GitHub project to demonstrate the progress claimed above. Please provide links for individual files or directories pointing to the work you have done over the past 2 weeks. This will help TAs and peer reviewers efficiently assess factual accuracy.
- Datasets would show metadata, followed by a sample of data items. Software platforms would show documentation, followed by a sample demo (Hello World). Your own code would show a code listing, followed by execution results.
- Add screenshots as necessary.

Repo: <https://github.com/NB670/ConstractLens>

This report contains the majority of the project planning claimed to have been done in the match section of this checkpoint. The datasets that were identified for use in this project were CUAD and ACORD.

Skill Learning Report

Mention each skill, followed by 1-2 lines on how you are making progress towards learning the skills or implementing them in your project, or a 3-4 line explanation on how you have learnt a particular skill.

Legal Document Understanding

Learning how commercial documents are structured and how important clauses are currently identified and analyzed.

Information Extraction

This is for learning how to take unstructured data and convert it into a format easy for downstream tasks and LLMs to understand.

Semantic retrieval

This skill is to learn how text can be vectorized and represented then compared for analysis and other tasks.

NLP in the legal domain

This skill is to learn how to take legal dataset information and use models to create accurate classification and risk analysis.

Local AI systems

How can quantized local models and MCP servers be used to create something privacy preserving but still accurate.

Self-Evaluation

1. Plan: 110

- a. Goes beyond yet another by focusing on converting unstructured text into something that can be molded to any task. Also takes privacy into consideration with local models which can be tuned to specialize in the task at hand.

2. Match: 110

- a. Everything claimed to have been done with planning was done but also extensive research was done on competitors and ensuring the plan was not just an empty demo or yet another project.

3. Factual: 100

- a. This document serves as evidence of project planning and the github link is in it too

LLM Feedback:

1. Project Plan (Plan): 112 / 120

The project has a well-defined enterprise use case, strong datasets, and meaningful backend functionality beyond a standard chatbot interface.

2. Progress Report (Match): 118 / 120

The completed work aligns with the goals of the initial planning checkpoint and establishes a clear roadmap for future development.

3. Supporting Evidence (Factual): 95 / 100

The proposal includes datasets, architecture, milestones, and repository documentation, with implementation evidence planned for future checkpoints.

Actionable Suggestions:

1. Prioritize implementation of contract ingestion and clause classification before developing advanced LLM features.
2. Establish baseline evaluation metrics using the CUAD dataset early in development to track progress throughout the project.
3. Focus on contract comparison and risk analysis as key differentiating features beyond standard document question answering.
4. Maintain a modular architecture so clause classification, retrieval, risk analysis, and LLM components can be independently evaluated and improved.
5. Treat fine-tuning, advanced retrieval methods, and multimodal contract understanding as stretch goals after the core platform is operational.